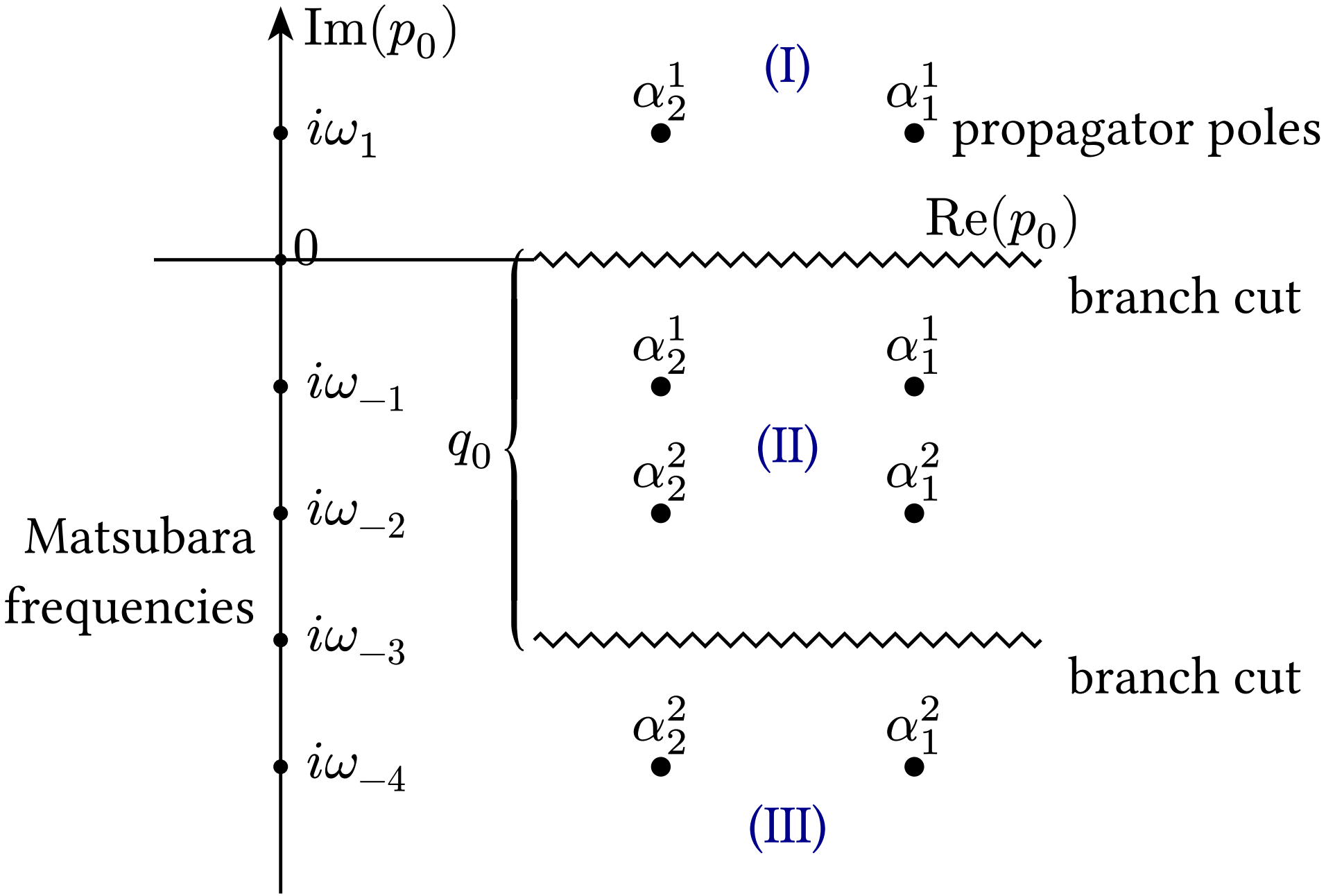
$\mathrm{Im}(p_0)$
$i\omega_1$
0
$i\omega_{-1}$
$i\omega_{-2}$
$i\omega_{-3}$
$i\omega_{-4}$
Matsubara frequencies
q_0
(I)
(II)
(III)
α_2^1
α_1^1
α_2^2
α_1^2
propagator poles
$\mathrm{Re}(p_0)$
branch cut
branch cut